

Project Problem Statement

Project Title

AI Programming Lab Mentor

Problem Statement

Programming is a fundamental skill for students pursuing computer science and related disciplines. However, many beginners face difficulties in understanding programming concepts, interpreting compiler errors, debugging code, and developing problem-solving skills. During laboratory sessions, instructors often need to assist many students simultaneously, making it difficult to provide immediate and personalized guidance to everyone.

As a result, students experience delays in resolving programming issues, lose confidence, and may struggle to complete laboratory exercises independently.

There is a need for an intelligent virtual mentor that can provide instant assistance, explain programming concepts in simple language, identify common coding mistakes, suggest possible solutions, and generate practice questions to improve students' programming skills.

The proposed **AI Programming Lab Mentor** leverages Artificial Intelligence to provide real-time learning support, helping students understand programming concepts more effectively while reducing the workload of instructors during laboratory sessions.

Proposed Solution

The AI Programming Lab Mentor is an AI-powered educational assistant designed to support students throughout their programming learning journey. The system provides:

- Programming concept explanations
- Code analysis and debugging assistance
- Algorithm and pseudocode guidance
- Practice question and quiz generation
- Personalized learning recommendations
- Instant responses to students' programming queries

By integrating Generative AI technologies, the proposed system enhances the learning experience, promotes self-paced learning, and improves students' confidence in programming.

Expected Outcome

- Improved programming knowledge among students.
- Faster identification and correction of programming errors.
- Increased student engagement during laboratory sessions.

- Reduced dependency on instructors for basic programming queries.
- Enhanced overall learning efficiency through AI-assisted mentoring.

Domain

Artificial Intelligence (AI), Generative AI, Education Technology (EdTech)

Target Users

- Undergraduate students
- Programming beginners
- Faculty members conducting programming laboratories
- Educational institutions